Final Project Report

- Waking up the family -

Team 2
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Introduction

The goal of this game is 'Waking up the family and turn off the alarm'. The player has to assist family members in waking up on time for their daily routines such as work or school, by finding them and gathering each necessary items based on the stereo sound effects. The game is played in the third-person view.

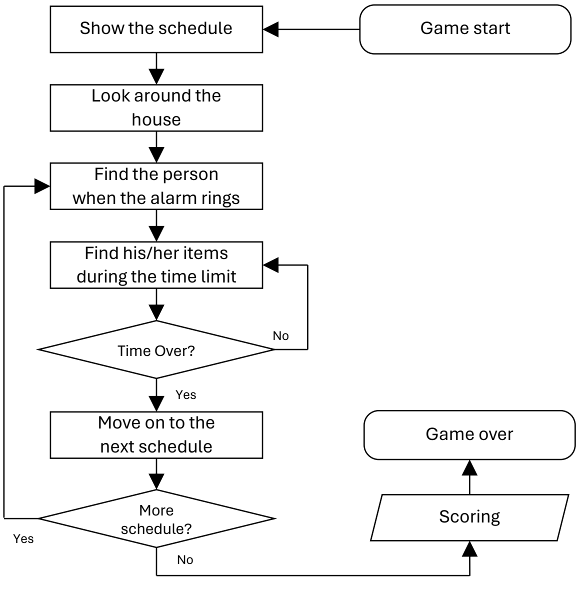
User Interface

- We created all the UI in the game ourselves and designed it overall , including (2) how to play the game for a player, (3) introducing all family members, (4) selecting levels, (5) notifying what tasks to do as each schedule begins, and (6) marking scores.

Game Development

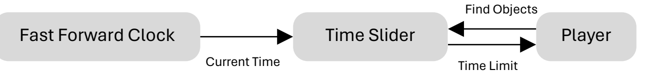
- **Game Sequence**

- When the game starts, the player can look around the house and find the family members and items following the schedules. When the schedules are over in order, there is a scoring and game will be over.



- **Time System**

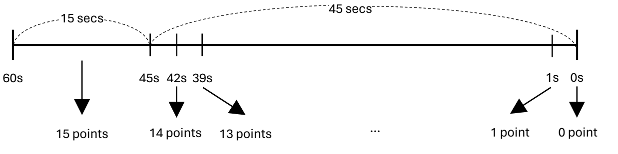
- In time system, 1 minute in game is equal to 1 second in real world. If the schedule starts, the time limit is reduced for 1 hour in game. During the time limit, player has to move around to do tasks.



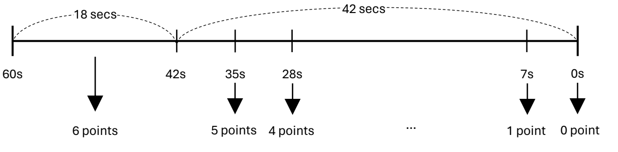
- **Scoring System**

- In scoring system, the total score of one character is 15 points, and the total score of one item is 6 points.
- We created the score equations, and each score depends on the remained time in Time Slider.

① Character



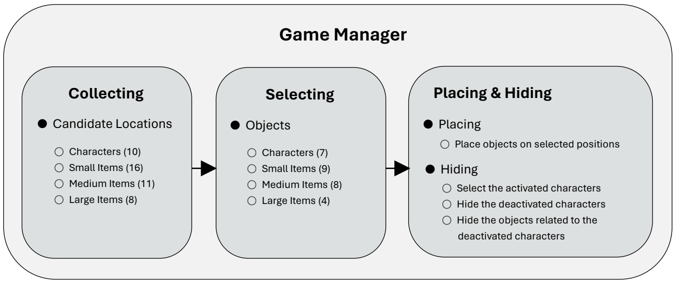
② Item



- Character score = 15 – (currentTime / 60 - 15) / 3
- Item score = 6 – (currentTime / 60 – 18) / 7

- **Game Positioning System**

- In game positioning system, we have already designated the candidates for the locations where the objects will be appeared.
- The game manager will select specific positions to be spawned randomly.

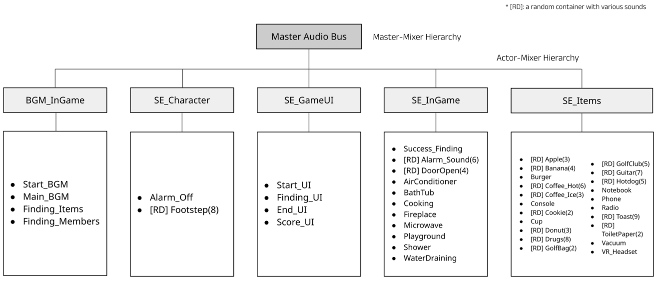


Audio System

- **Overall Sound Components**

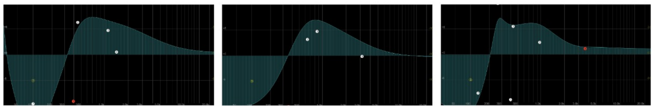
Sound	Source	Explanation
1) Alarm	**Record**	Cell phone alarm sound (iPhone, Galaxy)
2) Items	**Online**	The sounds of randomly placed items to find
3) Ambience	**Online**	The sound of house ambience
4) Game UI	**Online**	Sound effects and BGM of UI generated by interaction within the game

- **Mixer Hierarchy in Wwise**

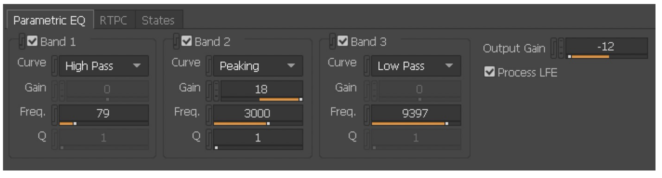


- **Alarm Sound**

- Recording: stereo sounds with XY technique
- Mixing in Reaper (EQ): To remove noise and make it sound as close as possible



- Mixing in Wwise (EQ): Even though recorded it, it sounded like the original sound effect applied the Band Pass like the sound from the phone



- **Item Sound**

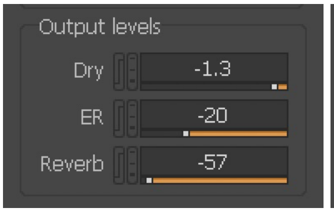
- A log graph with a specific distance is used for items to be found to minimize the player's confusion in distinguishing between sounds on the 1st and 2nd floors and to better identify the exact location of the sound.

[Figure 1] Sound of the items to find using a log graph

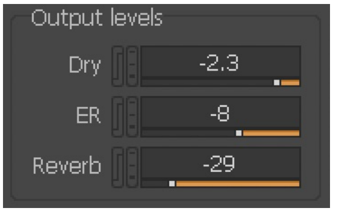
[Figure 2] Sound of the other ambience using an exponential graph

- A reverb applied to all items in Wwise, simulating indoor sounds.
- Guitar sounds and shouts of "Hotdog" enhanced with additional reverb.
- Popping sounds such as wrapping foods or can-opening handled with a compressor.
- In machine noises, an EQ (Band Pass) was used to achieve a mechanical tone,
- In doughnut sounds, a lower the pitch and the Low Pass Filter was applied.

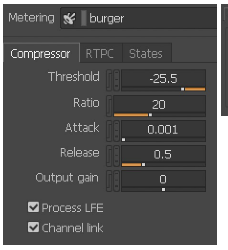
① Default Reverb Setting



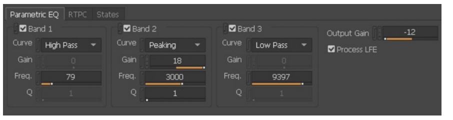
② Guitar Reverb Setting



③ Burger Compressor

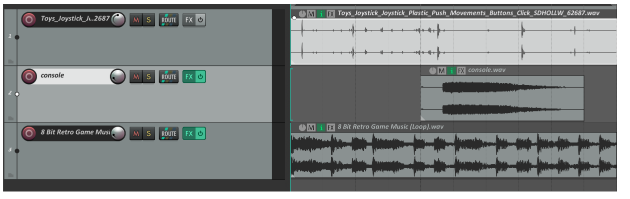


④ VR Headset EQ



⑤ Additional mixing

In e.g. console sound, a sound combination of console key-pressing with 8-bits game BGM and sound effects



- **Ambience Sound**

The sound of things in the house to disturb the player.

- 1F : Frying(High Pass), Microwave(Low Pass), Draining water(Low Pass), Shower booth (Reverb)
- 2F : Fireplace (High Pass, Pitch Down), Bathtub (Reverb), Air conditioner (Reverb)
- Outside : Playground (Reverb)

- **Game UI Sound**

Basic game interaction and success feedback sound

Home	**BGM**	Creating an interesting atmosphere
	Sound Effects	Feedback from the click-on interaction
In-game	**BGM**	The creation of a sense of urgency It increases the difficulty of finding things However, adjust it to sound subtle rather than clear (based on wearing the headset)
	Sound Effects	•Instruction to move around freely at first •Instruction to find people and things •Success feedback discovering people and Items •Final completion pop-up feedback

Limitation

- Instead of implementing a 3-stage level system (Easy, Medium, Hard), only one stage was implemented under the assumption that it is a Medium Level.
- Without distinguishing walls, it is based only on distance; therefore, sounds from the first floor are clearly heard on the second floor, and there is an issue where distinctions between rooms are not made.
- The score is not based on the distance in Wwise but is determined by how quickly one finds the source, with higher scores awarded for faster findings.